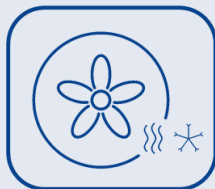


Renovation Scenarios



Passive System Scenarios

Envelope renovations that are based on the Turkish (TS825) standard were applied for the building envelopes.



Active System Scenarios

Heat pump systems were installed for both space heating and cooling.



Renewable System Scenarios

The rooftop PV electricity generation was calculated at the building level.

Key Performance Indicators (KPI)s

Environmental Indicators

Environmental KPIs indicate how much a building contributes to environmental impacts through its energy consumption.



Heating and cooling energy consumption refer to the total amount of energy required to maintain comfortable indoor air temperatures. These are expressed in kWh/m²



PV-based electricity supply quantifies the amount of clean electricity generated through photovoltaic systems.



CO₂ reduction represents the decrease in operational CO₂ emissions achieved through energy renovations, relative to a baseline scenario. CO₂ reduction is commonly reported in tonnes of CO₂ per year.

Economic Indicators

Economic KPIs provide insight into the financial viability and investment performance of the proposed measures.



Return on Investment (ROI) compares the economic savings to the upfront cost; a higher ROI means better financial efficiency.



The discounted payback period (DPBP) indicates the number of years required to recover the initial investment through discounted cash inflows— here, shorter periods are preferred.



Net Present Value (NPV) is the present value of expected cash flows minus the initial cost; a positive NPV means the measure creates value at the chosen discount rate.



Abatement cost is the cost to avoid one tonne of emitted carbon dioxide; lower or negative values indicate more cost-effective emission reductions. Used together, these metrics give a clear view of profitability, speed of payback, and cost-effectiveness.

Thermal Comfort Indicator

Thermal comfort indicators capture indoor environmental quality from the occupants' perspective.



The indoor overheating degree (IOD) combines the intensity and duration of temperatures above a maximum threshold, and is highly relevant for climate-resilience evaluation. We employ the IOD as a measure of cumulative thermal discomfort as well as the threshold-based pass/fail criteria set by the Chartered Institution of Building Services Engineers standards.